Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose **RandomTree** -K 0 -M 1.0 -V 0.001 -S 1

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 5
- Percentage split — % 66

More options...

(Nom) class

Start | Stop

Result list (right-click for options)

- 13:22:11 - trees.J48
- 13:38:29 - functions.SMO
- 14:10:11 - bayes.NaiveBayes
- 14:10:25 - functions.Logistic
- 14:12:36 - trees.RandomTree

Classifier output

```
|   |   |   |   |   pedi >= 1.41 : tested_negative (3/0)

Size of the tree : 271

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         557               72.526  %
Incorrectly Classified Instances       211               27.474  %
Kappa statistic                          0.4021
Mean absolute error                      0.2747
Root mean squared error                  0.5242
Relative absolute error                 60.4458 %
Root relative squared error            109.9681 %
Total Number of Instances              768

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.776    0.369    0.797      0.776    0.786      0.402    0.703     0.764     tested_negative
                 0.631    0.224    0.601      0.631    0.616      0.402    0.703     0.508     tested_positive
Weighted Avg.    0.725    0.319    0.729      0.725    0.727      0.402    0.703     0.675

=== Confusion Matrix ===

   a   b   <-- classified as
 388 112 |   a = tested_negative
  99 169 |   b = tested_positive
```

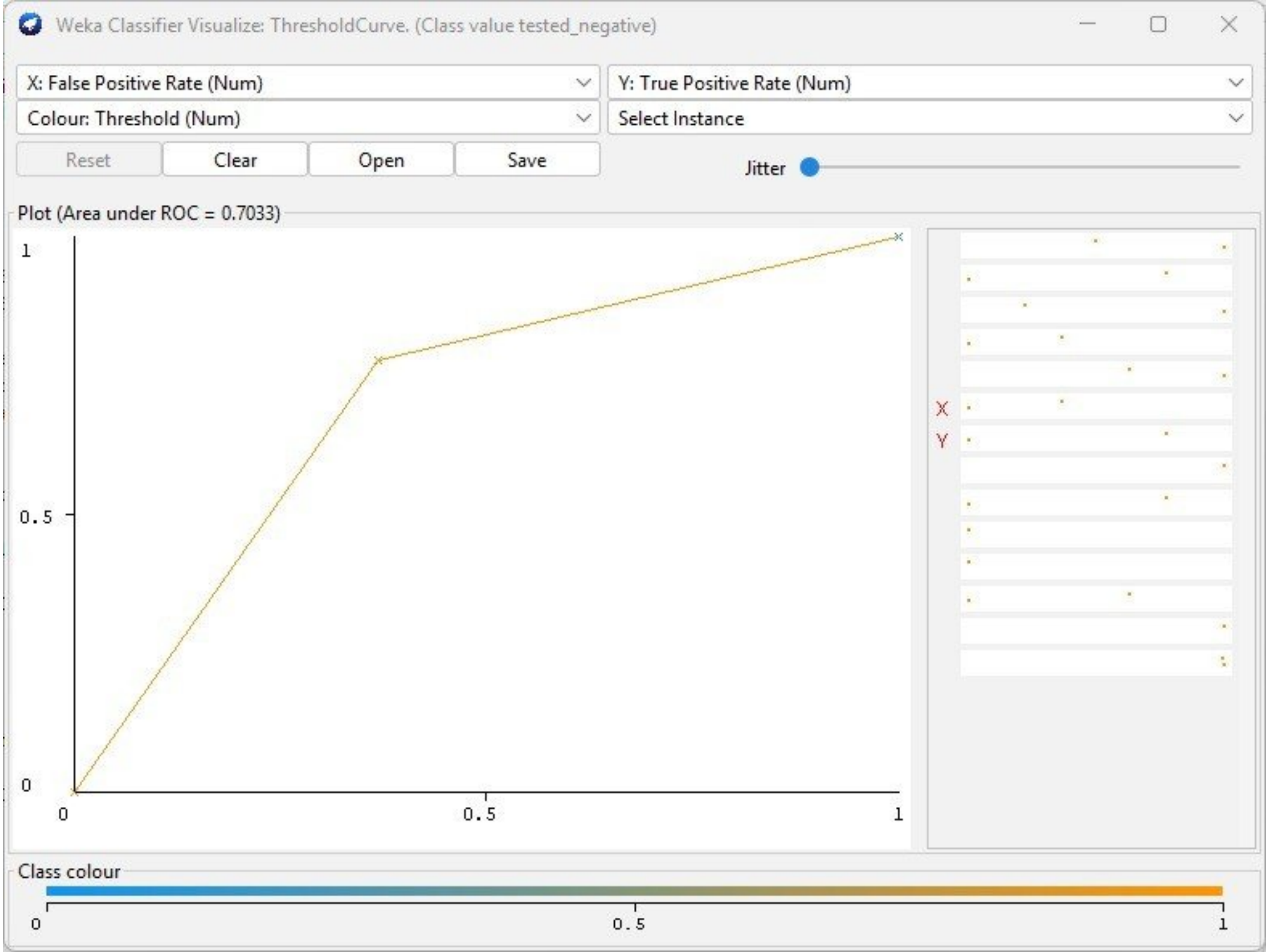
Activate Windows
Go to Settings to activate Windows.

Status

OK

Log

x 0


Weka Classifier Visualize: ThresholdCurve. (Class value tested_negative)
X: False Positive Rate (Num)
Y: True Positive Rate (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot (Area under ROC = 0.7033)
1
0.5
0
0
0.5
1
X
Y
Class colour
0
0.5
1

Weka Explorer

Preprocess | Classify | Cluster | Associate | Select attributes | Visualize

Classifier

Choose **RandomForest** -P 100 -I 100 -num-slots 1 -K 0 -M 1.0 -V 0.001 -S 1

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds 5
- Percentage split — % 66

More options...

(Nom) class

Start | Stop

Result list (right-click for options)

- 13:22:11 - trees.J48
- 13:38:29 - functions.SMO
- 14:10:11 - bayes.NaiveBayes
- 14:10:25 - functions.Logistic
- 14:12:36 - trees.RandomTree
- 14:13:16 - trees.RandomForest

Classifier output

```
Bagging with 100 iterations and base learner

weka.classifiers.trees.RandomTree -K 0 -M 1.0 -V 0.001 -S 1 -do-not-check-capabilities

Time taken to build model: 0.14 seconds

=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances         587               76.4323 %
Incorrectly Classified Instances       181               23.5677 %
Kappa statistic                          0.4698
Mean absolute error                      0.3115
Root mean squared error                  0.4038
Relative absolute error                 68.5386 %
Root relative squared error             84.7111 %
Total Number of Instances              768

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
               0.844     0.384     0.804      0.844    0.823      0.471    0.821     0.893     tested_negative
               0.616     0.156     0.679      0.616    0.646      0.471    0.821     0.671     tested_positive
Weighted Avg.  0.764     0.305     0.760      0.764    0.761      0.471    0.821     0.816

=== Confusion Matrix ===

   a   b   <-- classified as
 422  78 |   a = tested_negative
 103 165 |   b = tested_positive
```

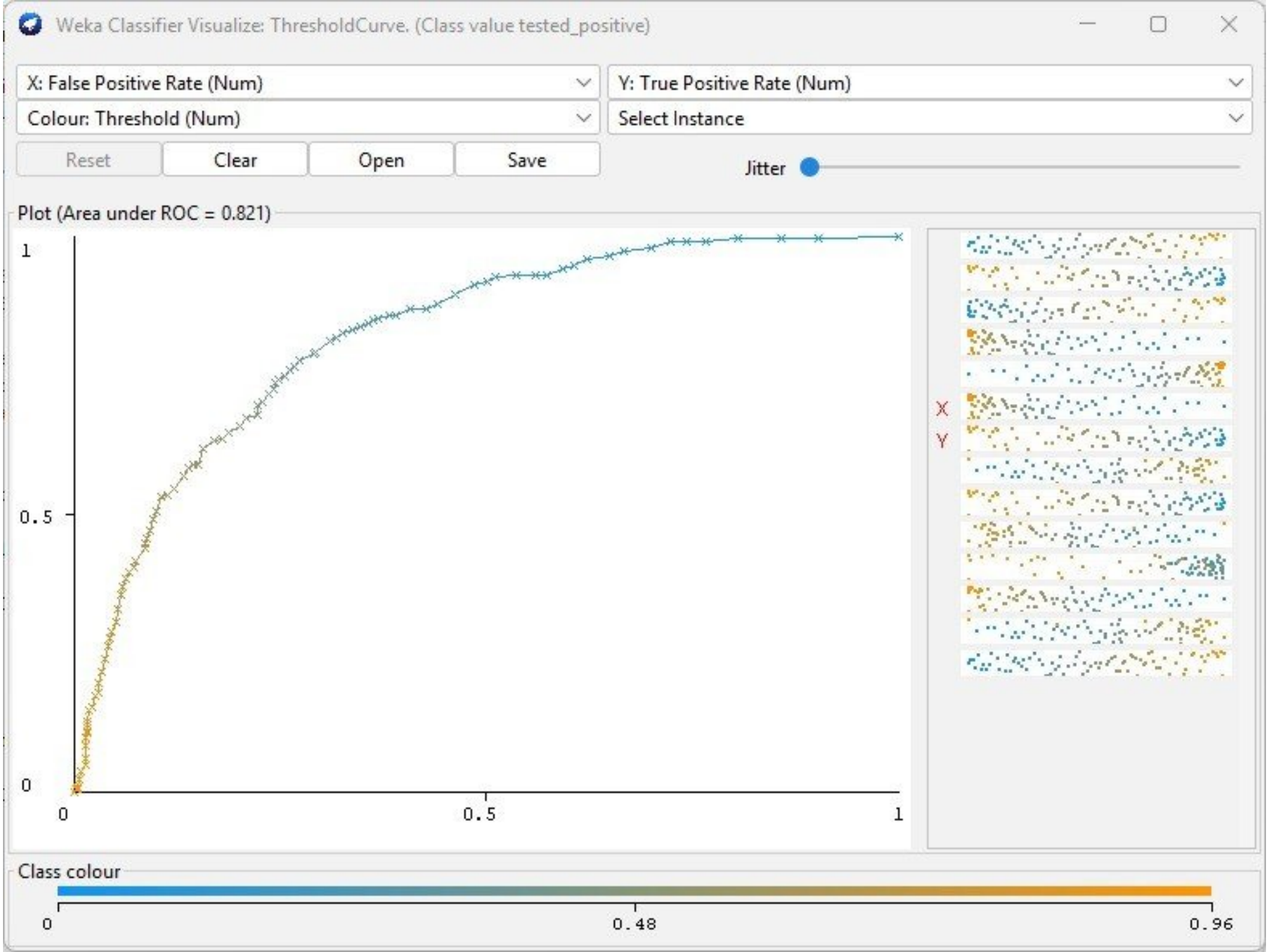
Activate Windows
Go to Settings to activate Windows.

Status

OK

Log

x 0


Weka Classifier Visualize: ThresholdCurve. (Class value tested_positive)
X: False Positive Rate (Num)
Y: True Positive Rate (Num)
Colour: Threshold (Num)
Select Instance
Reset
Clear
Open
Save
Jitter
Plot (Area under ROC = 0.821)
1
0.5
0
0
0.5
1
X
Y
Class colour
0
0.48
0.96